Driven oscillator - handout

Example PDF for the Interactive Presenter demo

A PDF on a slide can be scrolled and zoomed during the talk, so a handout, a paper or a data sheet can be shown as it is.

The model

A mass on a spring with damping, driven at frequency f, settles into oscillation at the drive frequency. Its amplitude depends on how close f is to the natural frequency f_0 and on the quality factor Q:

$$A(f) = 1 / \sqrt{((1 - (f/f_0)^2)^2 + (f / (f_0 Q))^2)}$$

What to notice

- At $f = f_0$ the amplitude is Q times the static one.
- The peak's width is about f_0/Q: a high Q means a sharp, tall resonance.
- Far above resonance the mass hardly moves.

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